🔑 **Answer Key — Teacher Copy · Lesson 7-4**

MSTAR PRACTICE WORKSHEET

Lesson 7-4: Compare and Order Integers and Rational Numbers

Grade 6 Mathematics · Standard 6.NOS.8

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: A

Which statement is TRUE?

- **A** -5 < -2
- **B** -4 > 0
- **C** -3 > -1
- **D** -1 < -6

-5 is farther to the left on the number line than -2, so -5 < -2 is true.

Question 2 SELECTED RESPONSE — CORRECT: A

Which integer is the LEAST: -3, 5, -8, 2, 0?

- **A** -8
- **B** -3
- **C** 0
- **D** 5

-8 is farthest to the left on the number line, making it the least (smallest) integer.

Question 3 SELECTED RESPONSE — CORRECT: B

What is |−8|?

A 0

B 8

C −8

D −16

Absolute value is distance from 0, never negative (zero is allowed). |−8| = 8.

Question 4 SELECTED RESPONSE — CORRECT: C

Three cities have temperatures of -12°F, -5°F, and 3°F. Which city is warmest?

A -5°F

B -12°F

C 3°F

D They are equal

3°F is the greatest because it is farthest to the right on the number line. The order from coldest to warmest is -12, -5, 3.

Question 5 SELECTED RESPONSE — CORRECT: A

Four spots: -3 m, 2 m, -7 m, 0 m. Order them from least to greatest.

A -7, -3, 0, 2

B -7, -3, 2, 0

C -3, -7, 0, 2

D 2, 0, -3, -7

The deepest spot is the cave at -7 (farthest left), then the tide pool -3, then the boat deck 0, then the flag at 2. Least to greatest: -7, -3, 0, 2.

Question 6 SELECTED RESPONSE — CORRECT: D

Three buried chests sit at -9 m, -4 m, and -15 m below sea level. Which chest is at the GREATEST elevation (closest to the surface)?

A -9 m

B -15 m

C They are equal

D -4 m

Closer to zero means greater for negatives. -4 is only 4 below the surface, so -4 is the greatest of -9, -4, and -15.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: C

Order these numbers from LEAST to GREATEST: −2, 1.5, −4.5, 0, 3.

A −2, −4.5, 0, 1.5, 3

B 3, 1.5, 0, −2, −4.5

C −4.5, −2, 0, 1.5, 3

D −4.5, −2, 0, 3, 1.5

Reading a number line from left to right gives −4.5, −2, 0, 1.5, 3.

- **A:** The two negatives are reversed. Negative 4.5 sits further left than negative 2, so it comes first.
- **B:** That is greatest to LEAST. The question asks for least to greatest.
- **D:** The last two are reversed. 1.5 is smaller than 3.

Part B — correct answer: B

Which statement correctly compares −4.5 and −2?

A −4.5 > −2, because 4.5 > 2.

B −4.5 < −2, because −4.5 lies further left on the number line.

C They are equal in value.

D −4.5 > −2, because it is further from zero.

Left is less. −4.5 sits to the left of −2, so −4.5 is the smaller number even though its absolute value is larger.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, C, E

Select ALL comparisons that are TRUE:

- **A** $-7 < -2$
- **B** $0 > -1$
- **C** $-0.5 > -1.5$
- **D** $-10 > -3$
- **E** $2.5 > -9$

A, B, C, and E are true. D is false — −10 lies further left than −3, so $-10 < -3$.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A student says: 'Negative numbers are always less than positive numbers, so -100 is less than 1.' Is this correct? Now the student says: '-100 is less than -1 because 100 is greater than 1.' Is THIS correct? Explain the difference.

Model answer: The first statement is true: -100 is negative and 1 is positive, so -100 sits left of 1 on the number line and is less. The second answer, $-100 < -1$, is also true, but the reason given is wrong — it is not because $100 > 1$, it is because -100 lies 100 units below zero while -1 lies only 1 unit below, putting -100 farther left.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.